

HEARSAY: The Audio Authentication Challenge

Welcome to HEARSAY

Hello, and welcome to your first assignment with the National Security Agency's Multimedia Forensics and Authentication team. Three seconds. That's all a modern voice-cloning system needs to impersonate a head of state, a battlefield commander, or your company's CEO. A single fabricated audio clip can move markets, empty a bank account, redirect troops, or end a career before anyone thinks to ask *"is that real?"*

The stakes have never been higher. Your adversaries are already deploying synthetic-voice systems at scale. The question isn't whether the fake call will come — it's whether we'll know it's fake when it does.

Your mission: Build an AI system that can autonomously analyze an audio file and determine whether it is real or synthesized. Real audio clips might be augmented with noise and other perturbations.

The Challenge

Develop an intelligent audio authentication framework that can:

- Accept any audio file as input (WAV, MP3, M4A, MP4 audio, OGG, etc...)
- Automatically orchestrate multiple forensic analysis techniques which may include audio, spectrograms, signal processing and file metadata.
- Output a synthetic probability score (0.0 to 1.0), 1.0 means synthetic.
- You will receive a training dataset along with a separate, held-out test set for evaluation.
- You can record your own audio in various scenarios (e.g. cartoonish voices, rapid speaking) if desired to gain confidence or skepticism in your techniques.
- Clips will be at least 2 seconds and in English.
- Provide explainable results: what techniques were used to determine a clip was synthetic. Tell us what worked and what had no effect.

You have 36 hours. Cloud services, open-source LLMs, LLM resources provided by HexLabs or your own university resources, speech models, and signal-processing toolkits are all fair game. How you combine them is up to you.

What Makes This Hard

State-of-the-art systems can:

- **Clone a voice from as little as 3 seconds of reference audio** (zero-shot TTS)
- **Convert one speaker's voice into another's in real time** while preserving cadence and emotion
- **Insert or replace individual words** in an otherwise-genuine recording
- **Fabricate background acoustics** to place a speaker in a room they were never in or indoor vs. outdoor, added electrical background, sirens, or birdsong.
- **Launder artifacts through re-encoding**, phone-quality downsampling, or added noise to defeat naive detectors.
- **Spoof container metadata** so the file claims a device and timestamp that never produced it. Modify the MAC (Modified, Accessed, Created).

A single detector won't cut it. You'll need to think like the forger *and* the forensic analyst: What inconsistencies did the AI miss? What statistical fingerprints did it leave behind? Does the metadata match up?

Test Dataset

NSA will provide a curated, labeled dataset that may span:

- **Bona fide speech** — studio, smartphone, telephony, and field recordings across codecs and bitrates
- **Fully synthetic speech** — modern zero-shot TTS and neural vocoders
- **Voice-converted speech** — real source speaker, cloned target identity
- **Replay attacks** — a recording of a playback

- **Scene manipulation** — authentic voice, fabricated or replaced ambient background
- **Laundered fakes** — synthetics passed through transcoding, band-limiting, or noise to hide artifacts
- **Metadata-spoofed containers**

The sample training dataset contains labeled ground truth: synthetic (spoof) vs. real (bonafide).

A sample test set is also provided where labels are held out and used for system evaluation to determine the detection performance score as part of the scoring rubric. Submit the output tab-delimited file (.tsv) for this sample test set.

Note: To help each team refine their approach, NSA will offer a **one-time** review of each team's draft tab-delimited file to better assess their current performance snapshot. This is optional but recommended prior to official submission so there's sufficient time for the team to address any gaps.

Scoring Rubric

1. Detection Performance (60%)

- Synthetic probability score (0.0 to 1.0)

2. Forensic Diversity and Breadth of Detector Types (20%)

- Points for each distinct technique your system actually leverages
- Container, digital file forensics & metadata analysis (encoder tags, timestamps, codec chain, MAC (Modified, Accessed, Created))
- Spectral / frequency-domain analysis (spectrograms, band-limiting, vocoder harmonics)
- Prosody & phonetics (breath, pause, coarticulation, pitch-contour naturalness, repeatability, monotonicity, emotion)
- Acoustic-environment consistency (ENF mains-hum)
- Compression forensics (double-encoding, transcoding artifacts)

- Speaker-embedding consistency (does the voice drift across the clip?)
- Deep-learning anti-spoofing detectors
- Splice / discontinuity detection (phase breaks, DC offset, background seams)
- Agentic orchestration bonus: Does your system choose which analyses to run based on file type, codec, initial findings, or confidence — rather than brute-forcing everything?

3. Documentation and Presentation (20%)

- How well is approach documented / communicated?
 - Can algorithm provide insight into why an audio is synthetic/real?
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Recommended Resources & Tools

You're free to use any open-source or commercial tools. Here are some battle-tested starting suggestions:

Signal processing & inspection

- librosa, torchaudio, scipy.signal, PyWavelets
- FFmpeg / ffprobe / MediaInfo — container, codec, and stream inspection
- SoX, Praat — acoustic analysis, pitch/formant extraction
- ExifTool, libExiv2 – Metadata Analysis

Anti-spoofing & deepfake detectors

- AASIST / RawNet2 / RawNet3 — end-to-end spoof detection
- wav2vec 2.0 / WavLM / Whisper encoders as front-ends for spoof classifiers
- ECAPA-TDNN / x-vectors — speaker embeddings for intra-clip consistency checks
- ASVspoof challenge baselines (LFCC-GMM, LCNN)

Datasets for training / calibration

- ASVspoof 2019 LA / 2021 LA+DF / ASVspoof 5
- WaveFake, In-the-Wild, MLAAD — multi-vocoder synthetic speech
- ADD Challenge (Audio Deepfake Detection) sets — includes partial fakes
- VCTK / LibriSpeech / VOiCES — bona fide references

Agentic frameworks

- LangChain / LangGraph — multi-step forensic workflows with LLM reasoning

- CrewAI — multi-agent (e.g., metadata agent, spectrum agent, embedding agent, adjudicator)
- HuggingFace Transformers / SpeechBrain — model deployment

References

- ASVspoof & ADD challenge overview papers (Interspeech / ICASSP)
 - DARPA SemaFor program — semantic forensics
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Submission Requirements

Your submission must include:

1. Source code
Provide a GitHub/GitLab repository containing your full implementation. The repository must include a **README** that clearly documents your approach, system architecture/design, and instructions for running your solution.
2. Docker container
Supply a Docker image that allows judges to run your system for reproducibility. The container should execute inference on the provided test set without major configuration changes.
3. Prediction file (Tab-Delimited file)
Submit a tab-delimited output file (.tsv) generated from the sample test set. The file must include a header row with the following columns:
 - filename: The name of the audio file being evaluated, including its extension (string).
 - cm-score: The predicted probability that the audio is synthetic, expressed as a float from 0.0 to 1.0. Highest probability a file is synthetic is 1.0.

Example Format:

```
teamName_predictions.tsv
filename  cm-score
file1.wav 0.80
file2.mp3 0.10
file3.m4a 0.40
file4.m4a 1.00
```

Why This Matters

This isn't hypothetical. The techniques you build this weekend could:

- Protect your own family once you know what today's voice-cloning scammers can do
- Flag a cloned-voice order before a unit acts on it
- Authenticate an intercepted call before it drives a national-level assessment
- Kill a CEO-voice wire-fraud attempt in the first 30 seconds
- Determine whether a viral "leaked recording" of a foreign leader is real before it shapes policy
- Stand up in a courtroom when the evidence is a voice, not a document

The fake voices are already out there. Your detector needs to be too.

Questions?

Join NSA's discord channel through HexLabs or visit the NSA booth where technical mentors will be available for questions.

Good luck. The nation's information integrity depends on your ingenuity.